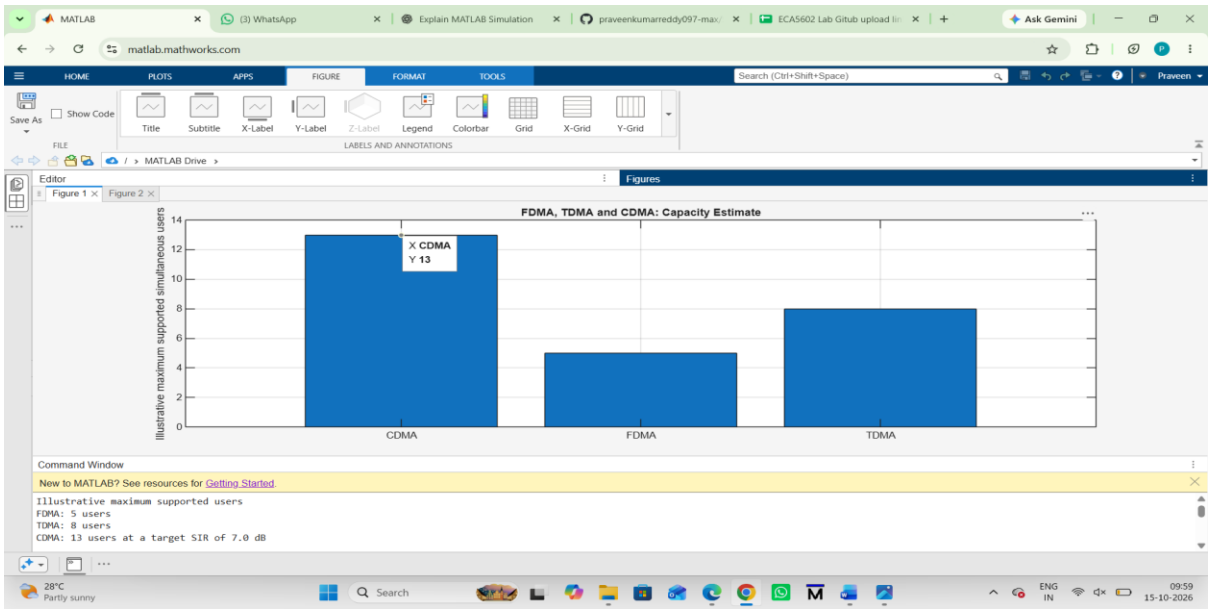
Experiment Number: 1 Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

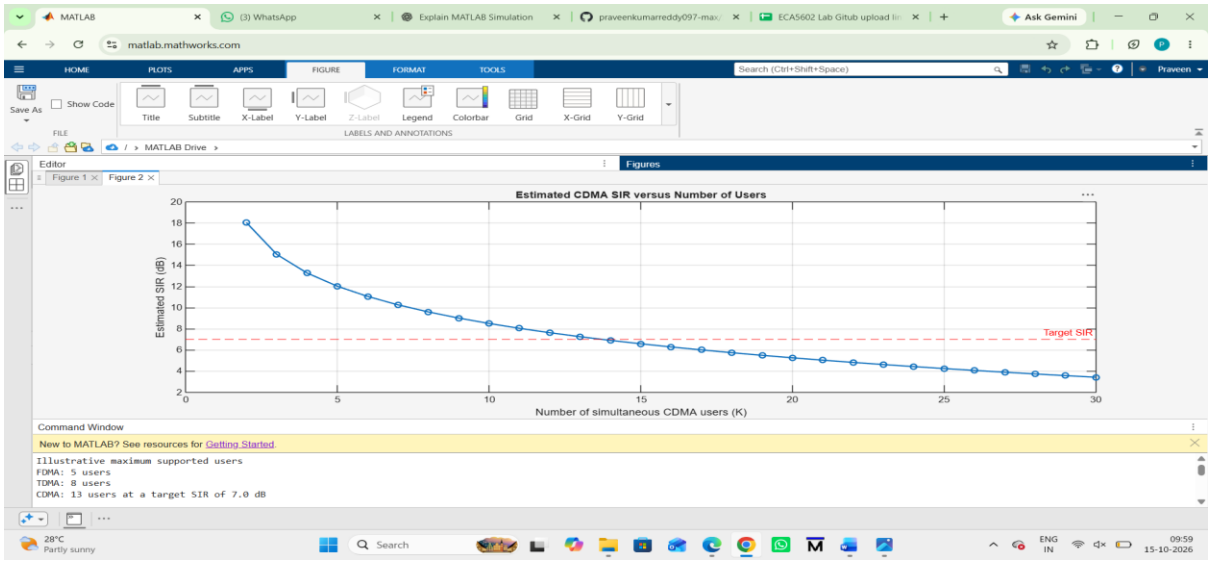
PROGRAM:

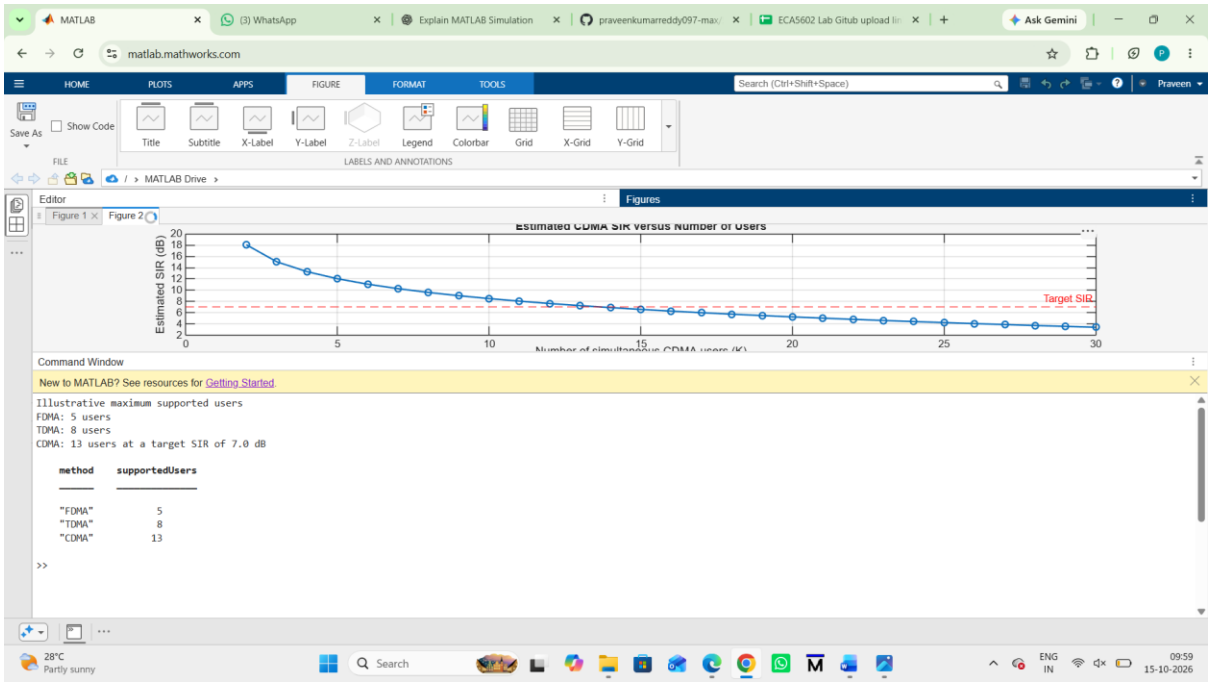
```
clear;
clc;
close all;
%% 1. Illustrative system parameters
totalBandwidthHz = 1e6;   % Total bandwidth: 1 MHz
fdmaChannelHz = 200e3;    % FDMA channel width: 200 kHz
tdmaSlots = 8;            % TDMA slots per frame
processingGain = 64;      % CDMA processing gain
targetSIRdB = 7;          % Minimum acceptable CDMA SIR in dB
%% 2. Estimate supported users
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
% Simplified CDMA capacity estimate
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB);
%% 3. Display comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);
%% 4. Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
xlabel('Multiple access technique');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
%% 5. Estimate CDMA SIR as users increase
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
```

```
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

OUTPUT:

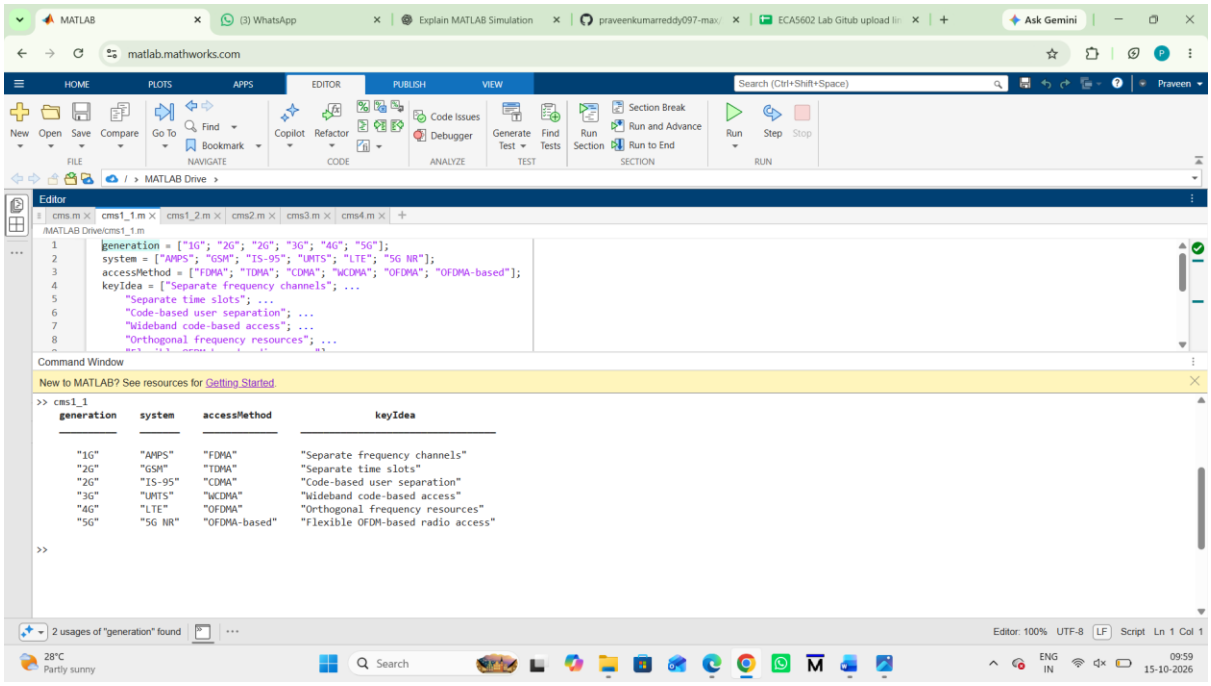



Estimated CDMA SIR versus Number of Users
Estimated SIR (dB)
Number of simultaneous CDMA users (K)
Target SIR
Command Window
New to MATLAB? See resources for Getting Started.
Illustrative maximum supported users
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB


Estimated SIR (dB)
Target SIR
Command Window
New to MATLAB? See resources for Getting Started.
Illustrative maximum supported users
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB
method supportedUsers
"FDMA" 5
"TDMA" 8
"CDMA" 13
>>

PROGRAM:

```
clear;
clc;
close all;
% Mobile communication generations
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
% Representative systems
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
% Multiple access methods
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; ...
                "OFDMA"; "OFDMA-based"];
% Key idea of each system
keyIdea = ["Analog voice communication"; ...
           "Digital voice using time slots"; ...
           "Digital voice using spread-spectrum codes"; ...
           "Wideband code-division access"; ...
           "Orthogonal frequency-division access"; ...
           "Flexible OFDM-based radio access"];
% Create comparison table
evolutionTable = table( ...
    generation, system, accessMethod, keyIdea, ...
    'VariableNames', ...
    {'Generation', 'System', 'AccessMethod', 'KeyIdea'});
% Display table
disp('Evolution of Mobile Communication Systems');
disp(evolutionTable);
```



PROGRAM:

```
clear;
clc;
close all;
%% Create one figure with three diagrams
figure('Color', 'w', 'Name', 'Multiple Access Techniques');

tiledlayout(1, 3, 'TileSpacing', 'compact', ...
             'Padding', 'compact');
colors = lines(4);
%% 1. FDMA: Different frequency channels
nexttile;
hold on;
for u = 1:4
    rectangle('Position', [0, u-1, 1, 0.8], ...
        'FaceColor', colors(u,:), ...
        'EdgeColor', 'k');
    text(0.5, u-0.6, sprintf('User %d', u), ...
        'HorizontalAlignment', 'center');
end
xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca, 'YTick', []);
```

```
box on;
hold off;
%% 2. TDMA: Different time slots
nexttile;
hold on;

for u = 1:4
   rectangle('Position', [u-1, 0, 0.8, 1], ...
      'FaceColor', colors(u,:), ...
      'EdgeColor', 'k');
   text(u-0.6, 0.5, sprintf('User %d', u), ...
      'HorizontalAlignment', 'center');
end
xlim([0 4]);
ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca, 'YTick', []);
box on;
hold off;
%% 3. CDMA: Shared time-frequency resources
nexttile;
hold on;
% Illustrative overlapping user transmissions
for u = 1:4
   rectangle('Position', [0, 0, 1, 1], ...
      'FaceColor', colors(u,:), ...
      'FaceAlpha', 0.15, ...
      'EdgeColor', colors(u,:), ...
      'LineWidth', 1.5);
end
text(0.5, 0.62, 'Users 1-4 share', ...
   'HorizontalAlignment', 'center');

text(0.5, 0.40, 'band; different codes', ...
   'HorizontalAlignment', 'center');
xlim([0 1]);
ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca, 'YTick', []);
```

```
box on;
hold off;
%% Overall title
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

OUTPUT:

